

## DIRECT-TO-CHIP LIQUID COOLING

# Designing a 6 kW AI accelerator cooling tray

A model-based workflow linking cold-plate CFD, reduced-order models, manifold hydraulics, and pump/fault decisions.

|                       |                        |                              |
|-----------------------|------------------------|------------------------------|
| <b>6.0 kW</b>         | <b>9.16 L/min</b>      | <b>12.3°C margin</b>         |
| 8 × 750 W design load | Modeled pump/tray flow | Below the 85°C project limit |

## The engineering decision

Retain a 19.05 mm common header as the provisional baseline. At the nominal modeled operating point, the tray reaches 72.72°C estimated peak device temperature with 0.342% flow maldistribution. The result is a design-study prediction, not hardware qualification. [E01–E02]

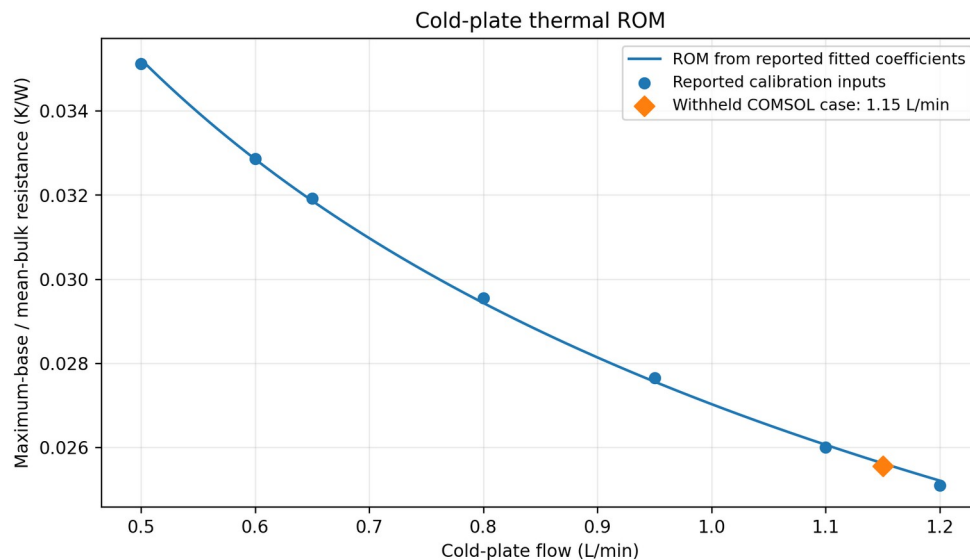
## From component physics to a system model

Screened cold-plate candidates in COMSOL; selected a 30-channel, 0.40 × 1.00 mm Candidate C design. Fitted pressure-drop and thermal-resistance ROMs, then integrated branch tubing, illustrative QD/fitting losses, and a scalable same-side manifold network. A representative pump curve determines actual modeled flow rather than prescribing the thermal target. [E10]

## Independent component-level CFD check

The 1.15 L/min, 750 W, 30°C COMSOL case was withheld from fitting. The extended calibration flow range is 0.50–1.20 L/min. [E03–E04]

| Output                   | COMSOL        | ROM           | Difference |
|--------------------------|---------------|---------------|------------|
| Pressure drop            | 11.258 kPa    | 11.260 kPa    | +0.0178%   |
| Thermal resistance       | 0.0255542 K/W | 0.0256179 K/W | +0.2493%   |
| Maximum base temperature | 53.870°C      | 53.908°C      | +0.0382°C  |



Reported fit and withheld CFD point; resistance is referenced to  $(T_{in} + T_{out})/2$ . ROM agreement with CFD does not establish hardware accuracy.

# Hydraulic limits can fail before temperature

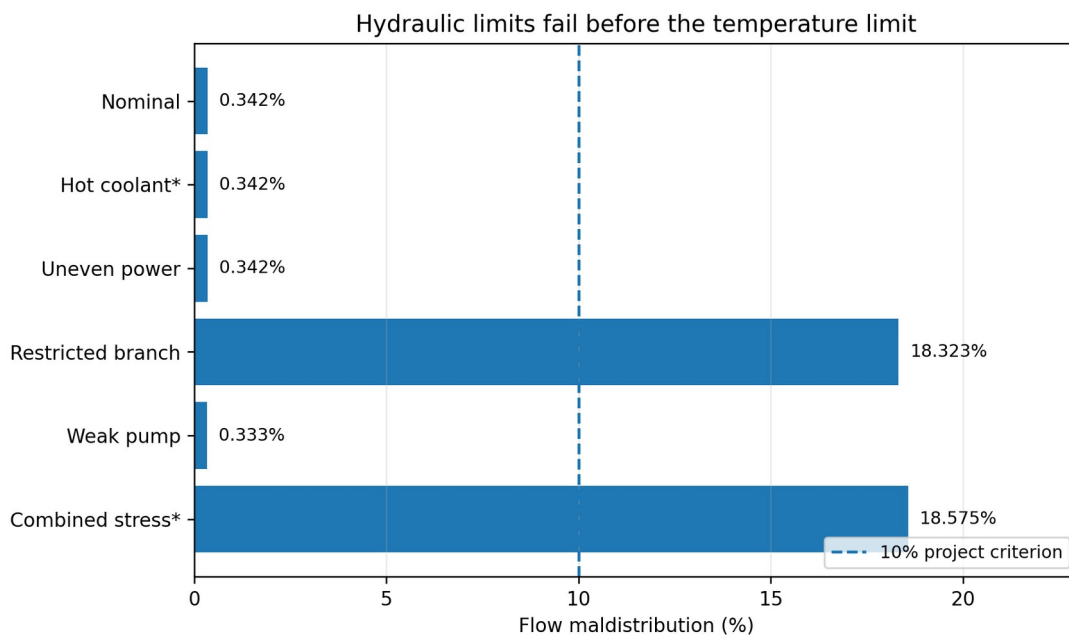
Six reported scenarios separate temperature compliance from flow and uniformity requirements. Nominal tray differential pressure is 11.44 kPa; the illustrative pump provides 6.02% flow margin above the fixed 8.64 L/min target. [E01–E02]

| Scenario          | Flow L/min | Mflow % | Tmax est. °C | Reported outcome        |
|-------------------|------------|---------|--------------|-------------------------|
| Nominal           | 9.1604     | 0.342   | 72.72        | Pass                    |
| Hot coolant*      | 9.1604     | 0.342   | 77.72        | Pass*                   |
| Uneven power      | 9.1604     | 0.342   | 72.72        | Pass                    |
| Restricted branch | 9.0632     | 18.323  | 75.05        | Fail: uniformity        |
| Weak pump         | 8.2056     | 0.333   | 74.10        | Fail: flow target       |
| Combined stress*  | 8.1016     | 18.575  | 81.67        | Fail: flow + uniformity |

\* 35°C inlet: constant-property sensitivity only, not additional inlet-temperature CFD validation. Mflow =  $100 \times (q_{\max} - q_{\min}) / q_{\text{mean}}$ . Status uses the adopted fixed-flow and 10% uniformity criteria.

## What changed the design judgment?

A 30% increase in one branch hydraulic characteristic raised maldistribution to 18.32% while the estimated device peak remained 75.05°C. The weak-pump case also remained below the temperature limit but missed the flow target. Temperature alone therefore would not identify every operating-requirement violation. [E01]



Fault results replotted from the supplied scenario table. No new CFD or full-system simulation was run to prepare this document.

## Scope of the recommendation

The pump curve, QD/fitting coefficients, and 0.025 K/W package-plus-TIM resistance are modeling assumptions. Pressure covers the tray, not a full CDU/facility loop. The 35°C cases are sensitivities; the complete manifold lacks independent CFD/hardware qualification. The 19.05 mm choice is provisional, not a global optimum or released mechanical assembly. [E10]

Evidence: E01 scenario table; E02 pump/tray summary; E03–E04 withheld component comparison; E10 model settings. Full source index and limitations accompany this document in the portfolio package.